FIELDWORK SPECIFICATION & TECHNICAL MANUAL

BuildCalc™

Construction ToolKit — Engineering Reference Manual

A rigorous, mathematically verified engineering design document and field reference manual for the **BuildCalc** application suite. Covering strongly-typed dimensional arithmetic, exact rational fraction engines, framing and board-foot geometry, modular CMU coordination, dry-stacked retaining wall mechanics, commercial ready-mix concrete batching standards, compound roofing rafter trigonometry, true-cut jack rafter offsets, and IRC residential stair dimensioning sentinels.

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- ***Section 2:*** *Rational Fraction Engine & Strongly Typed Physical Units*
- ***Section 3:*** *Materials & Structural Estimating (Concrete, Framing Lumber, Masonry)*
- ***Section 4:*** *Roofing Trigonometry & True-Cut Jack Rafter Schedule*
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1. Design System, Color Palette & Typography

BuildCalc is built around a field-first visual design system tailored for high-contrast readability under bright sunlight, glove-friendly touch ergonomics on active jobsites, and strict single-meaning semantic color assignment to prevent misinterpretation during critical structural installations.

1.1 Semantic Color System & Token Architecture

Every color role in the application is strictly enforced. Notably, **Brand Gold (#D4AF37)** has exactly one semantic meaning: **active selection and user interactivity**. It is never conflated with safety warnings or informational tags.

Color Token	Hex Code	Visual Role	Strict Jobsite Semantic Meaning
Brand Gold	`#D4AF37`	Accent / Active	Active focus, selected chip, interactive controls, app logo identity.
Technical Blue	`#38BDF8`	Annotation / Guide	Visual blueprint lines, guide dimension badges, stride comfort ratios.
Status Success	`#22C55E`	Safety / Pass	IRC building code pass, tolerance verified, within allowable limit.
Status Warning	`#F59E0B`	Alert / Boundary	Code violation, variance outside tolerance, caution advisory.
Background Dark	`#0E1217`	Canvas Base	Deep charcoal non-glare surface minimizing eye strain in outdoor glare.
Elevated Surface	`#161D26`	Card Container	Container surfaces, hero result modules, bottom sheet backgrounds.
Input Background	`#11161D`	Input Field	High-contrast recessed input background for numerical entry.
Text Primary	`#F5F7FA`	Primary Display	Primary calculation numbers, hero metrics, field titles.
Text Secondary	`#9CA3AF`	Labels / Units	Secondary units, field labels, dimensional fractions, helper subtitles.
Border Subtle	`#1E293B`	Dividers	1px structural card borders, tab bar dividing lines, table grid lines.

1.2 Canonical Typography Scale

The type hierarchy enforces tabular figure alignment so that numbers do not jitter during calculation updates, with distinct visual weight differentiation between whole dimensions, fractional denominators, and units.

Text Role	Size / Weight	Letter Spacing	Usage Example in Application
Primary Hero	52px / w800	-0.5px (Tabular)	**12' 7 3/8"**, **14 RISERS**, **4.25 yd³**
Screen Title	30px / w700	-0.3px	**Materials & Estimating**, **Roofing & Rafters**
Section Title	21px / w700	0.0px	**Standard Board Size**, **True-Cut Jack Schedule**
Card Title	18px / w700	0.0px	**Total Concrete Needed**, **Stair Steps Summary**
Result Unit	26px / w700	0.0px	**Total Volume**, **Board Feet**, **Line Length**
Input Label	17px / w500	0.1px	**Total Rise (Feet)**, **Common Rafter Run**
Input Value	28px / w600	0.0px (Tabular)	Editable numerical digits inside entry fields
Body / Explanatory	16px / w400	0.1px	Trade details expansion text, measuring instructions
Supporting Subtitle	14px / w400	0.0px	≈ 45 pre-mixed bags (80 lb / 36 kg) with 10% safety buffer
Micro / Badge	12px / w500	0.3px	**DIMENSION CHECK: PASS**, **16" O.C.**

2. Mathematical Engine & Exact Physical Quantities

Traditional building calculators suffer from floating-point accumulation errors (e.g., 0.1 + 0.2 = 0.30000000000000004) and multi-stage fractional rounding drift. BuildCalc eliminates this via a custom exact rational fraction arithmetic engine and strongly typed physical quantity wrappers.

2.1 Exact Rational Fraction Engine (Fraction Class)

The Fraction class represents all fractional tape measurements as an exact irreducible rational pair (numerator / denominator). It employs Euclidean Greatest Common Divisor (GCD) reduction on every arithmetic operation.

Euclidean GCD Algorithm:

```
gcd(a, b) = a if b == 0 else gcd(b, a mod b)
```

Least Common Multiple for Denominator Normalization:

```
lcm(a, b) = (|a * b|) / gcd(a, b)
```

Arithmetic Operators:

```
(p1 / q1) + (p2 / q2) = (p1 * (lcm / q1) + p2 * (lcm / q2)) / lcm(q1, q2)
(p1 / q1) - (p2 / q2) = (p1 * (lcm / q1) - p2 * (lcm / q2)) / lcm(q1, q2)
(p1 / q1) * (p2 / q2) = (p1 * p2) / (q1 * q2)
(p1 / q1) / (p2 / q2) = (p1 * q2) / (q1 * p2)
```

2.2 Single-Step Nearest Construction Snapping Proof

A critical bug in traditional framing apps is sequential multi-step rounding where 7.4038" is rounded to 7.404", then to 7.40", then to 7 7/16". BuildCalc enforces **single-step exact denominator resampling** in Fraction.fromDouble(value, maxDenominator):

```
whole = floor(|value|)
decimal = |value| - whole
fractionNumerator = round(decimal * maxDenominator)
totalNumerator = whole * maxDenominator + fractionNumerator
```

PROOF: Exact 1/16" Display for 96.25" Rise across 13 Steps

- *Exact theoretical riser: 96.25" / 13 =* ***7.40384615... inches***
- *Whole inches:* ***7"****; Decimal remainder:* ***0.40384615..."***
- *Scaled to 1/16" denominator: 0.40384615 * 16 =* ***6.461538...***
- *round(6.461538) =* ***6*** *(since 0.4615 < 0.5000)*
- *Raw fraction:* ***6/16"****; GCD reduction: gcd(6, 16) = 2 →* ***3/8"***
- *Distance to 3/8" (0.3750"): |7.403846 - 7.375000| =* ***0.0288"***
- *Distance to 7/16" (0.4375"): |7.403846 - 7.437500| =* ***0.0337"***
- ***Conclusion:*** *7 3/8" is the mathematically nearest 1/16" division. BuildCalc formats exactly as* ***7 3/8"****.*

2.3 Strongly Typed Physical Quantities & Exact Conversion Factors

All physical dimensions in memory are stored in standard base quantities (Inches for Length, Cubic Yards for Volume). Conversions use the official BIPM / NIST exact standard values without truncation.

Physical Dimension	Unit Symbol	Exact Conversion Standard	Mathematical Definition
Inch	`in / "`	1.0 inches (Base)	Exact base physical length unit
Foot	`ft / '`	12.0 inches	1 ft = 12 in
Yard	`yd`	36.0 inches	1 yd = 3 ft = 36 in
Millimeter	`mm`	1 in = 25.4 mm (Exact)	1 mm = 1.0 / 25.4 in
Centimeter	`cm`	1 in = 2.54 cm (Exact)	1 cm = 10.0 / 25.4 in
Meter	`m`	1 in = 0.0254 m (Exact)	1 m = 1000.0 / 25.4 in
Cubic Yard	`yd³`	1.0 yd^3 (Base Volume)	1 yd^3 = 27 ft^3 = 46,656 in^3
Cubic Foot	`ft³`	1 / 27.0 yd^3	27 ft^3 = 1 yd^3
Cubic Meter	`m³`	1.3079506193 yd^3	1 m^3 = $(1000/25.4/36)^3$ yd^3
Board Foot	`BF`	144.0 in^3 = 0.00235974 m^3	12" x 12" x 1" nominal lumber

3. Materials & Structural Estimating Engine

The Materials module provides estimation models across three essential jobsite trades: ready-mix and bagged concrete, dimensional framing lumber and board footage, and masonry brick and block walls.

3.1 Concrete Structural Form Geometry & Batch-Plant Norms

Volume calculations accurately convert mixed-dimensional inputs (length and width in feet/meters, thickness/depth in inches/centimeters) into exact cubic yards and cubic meters.

```
Rectangular Slab: V_cuft = L_ft * W_ft * (T_in / 12)
Continuous Trench Footing: V_cuft = L_ft * (W_in / 12) * (D_in / 12)
Cylindrical Column: V_cuft = π * ((D_in / 2) / 12)² * H_ft * Quantity
Foundation Wall: V_cuft = L_ft * H_ft * (T_in / 12)
Gross Volume with Waste: V_gross = V_net * (1.0 + (WastePercent / 100.0))
```

Commercial Ready-Mix Rounding (Quarter-Yard Increment):

Commercial concrete batch plants do not batch tenths of a yard. Orders are batched and dispatched in **quarter-yard (0.25 yd³)** increments. BuildCalc enforces this standard:

```
RecommendedOrderYards = ceil(V_gross_yards * 4.0) / 4.0
RecommendedOrderMeters = ceil(V_gross_meters * 4.0) / 4.0
```

Bagged Concrete Yield & Dead-Weight Standards:

- Pre-mixed 80-lb (36.3 kg) bag yield: **0.60 cu ft** per bag (45 bags per cubic yard).
- Hardened normal-weight concrete density: **145 lbs/cu ft** (~2,323 kg/m³) per ACI 318 structural dead load specification.

3.2 Framing Lumber & Board Feet Engine

A Board Foot is defined as the volumetric equivalent of a 1" thick, 12" wide, 1-foot long nominal board (144 cu in). **Board-foot volume must be calculated from nominal mill dimensions**, not planed actual dimensions.

```
BoardFeet = ((NominalThickness_in * NominalWidth_in * Length_ft) / 12.0) * Quantity
```

Nominal vs. Actual Finished Dimension Matrix:

Lumber Designation	Nominal Thickness	Nominal Width	Actual Finished Size	BF per 12-ft Piece
1x4	1.0"	4.0"	0.75" x 3.5" (19 x 89 mm)	`4.00 BF`
1x6	1.0"	6.0"	0.75" x 5.5" (19 x 140 mm)	`6.00 BF`
1x8	1.0"	8.0"	0.75" x 7.25" (19 x 184 mm)	`8.00 BF`
2x4	2.0"	4.0"	1.5" x 3.5" (38 x 89 mm)	`8.00 BF`
2x6	2.0"	6.0"	1.5" x 5.5" (38 x 140 mm)	`12.00 BF`
2x8	2.0"	8.0"	1.5" x 7.25" (38 x 184 mm)	`16.00 BF`
2x10	2.0"	10.0"	1.5" x 9.25" (38 x 235 mm)	`20.00 BF`
2x12	2.0"	12.0"	1.5" x 11.25" (38 x 286 mm)	`24.00 BF`
4x4	4.0"	4.0"	3.5" x 3.5" (89 x 89 mm)	`16.00 BF`
4x6	4.0"	6.0"	3.5" x 5.5" (89 x 140 mm)	`24.00 BF`
6x6	6.0"	6.0"	5.5" x 5.5" (140 x 140 mm)	`36.00 BF`

Wall Framing Stud Count Formula:

```
BaseStuds = ceil(WallLength_in / Spacing_in) + 1
CornerAllowance = Corners * 2 (Accounts for 3-stud California corners)
TeeAllowance = Intersections * 2 (Accounts for partition backing studs)
TotalStudsWithWaste = ceil((BaseStuds + CornerAllowance + TeeAllowance) * (1.0 +
WastePercent/100))
```

3.3 Masonry Block & Mortar Coordination

In modular masonry (ASTM C90), a standard 8x8x16 concrete block has manufactured dimensions of **7 5/8" x 7 5/8" x 15 5/8"**. When laid with a standard **3/8" (0.375")** mortar joint, it occupies exactly **8" x 8" x 16"** modular space (yielding the industry rule of thumb of **1.125 blocks per square foot**).

```
EffectiveLength = UnitLength + (isDryStacked ? 0.0 : MortarJointThickness)
EffectiveHeight = UnitHeight + (isDryStacked ? 0.0 : MortarJointThickness)
UnitsPerCourse = ceil(WallLength_in / EffectiveLength)
CoursesHigh = ceil(WallHeight_in / EffectiveHeight)
TotalUnits = ceil((UnitsPerCourse * CoursesHigh) * (1.0 + WastePercent/100))
```

• **Retaining Wall Blocks (12x4x8):** Laid dry-stacked with rear alignment lips. Zero mortar joint is added, and mortar bags calculate to 0.0.
• **Mortar Coverage Standard:** 32 standard hollow CMU blocks per 80-lb bag (face-shell bedding); 3.5 bags per 100 modular bricks.

4. Roofing & Rafter Cutting Trigonometry

Roof rafter solving requires exact compound angle trigonometry connecting plan angles (typically 45° at hip/valley intersections) with pitch slopes. BuildCalc provides mathematical line lengths, plumb and level cuts, and carpenter-friendly long-point cut marks.

4.1 Common Rafter Trigonometry

Roof Pitch Notation: Pitch = RiseIn12 / 12 (e.g. 4/12 or 6/12)

Pitch Angle: θ_pitch = arctan(RiseIn12 / 12.0) * (180.0 / π)

Unit Slope Multiplier: M_slope = sqrt(1.0 + (RiseIn12 / 12.0)²)

Theoretical Line Length: L_rafter = Run * M_slope

Plumb Cut Saw Angle: θ_plumb = 90.0° - θ_pitch (Set miter saw for ridge and plumb tail)

Seat Cut Level Angle: θ_seat = θ_pitch (Level notch resting flat on top wall plate)

Overhang Line Length: L_overhang = OverhangRun * M_slope

4.2 True-Cut Jack Rafter Geometry (Verified Model)

A jack rafter terminates against a diagonal hip or valley rafter. To fit tightly against the hip cheek, the theoretical centerline length must be shortened to account for half the hip thickness measured along the roof slope.

True-Cut Mathematical Deduction Derivation:

1. Plan Angle for Regular Hip: θ_plan = 45.0° (π / 4 radians)
2. Half-Hip Diagonal Thickness in Plan: Δ_plan = (HipThickness / 2.0) / sin(45.0°)
3. Sloped Shortening Deduction: Deduction_sloped = Δ_plan * M_slope
4. Centerline Progression for Jack #i: L_centerline(i) = (i * Spacing_in) * M_slope
5. Physical Cut Length to Long Point: L_cut(i) = max(0.0, L_centerline(i) - Deduction_sloped)
6. Compound Side-Cut Saw Bevel Angle: θ_saw = arctan(cos(θ_pitch)) * (180.0 / π)

Worked Example: 4/12 Pitch Jack Schedule (16" O.C., 1.5" 2x Hip Rafter)

- Pitch Angle: arctan(4/12) = **18.435°**
- Unit Slope Multiplier: sqrt(1 + (4/12)²) = sqrt(10/9) = **1.05409255**
- Plan Hip Half-Thickness: (1.5 / 2.0) / sin(45°) = 0.75 / 0.70710678 = **1.06066"**
- Sloped Hip Shortening Deduction: 1.06066 * 1.05409255 = **1.1180" (~1 1/8")**
- Compound Side-Cut Bevel Angle: arctan(cos(18.435°)) = arctan(0.94868) = **43.49° (~43.5°)**

Jack Number	O.C. Spacing	Theoretical Centerline	Hip Shortening	True-Cut Long Point	Compound Saw Bevel
Jack #1	16.0" (1' 4")	16.866" (1' 4 7/8")	1.118" (1 1/8")	**15.748" (1' 3 3/4")**	43.5°
Jack #2	32.0" (2' 8")	33.731" (2' 9 3/4")	1.118" (1 1/8")	**32.613" (2' 8 5/8")**	43.5°
Jack #3	48.0" (4' 0")	50.596" (4' 2 5/8")	1.118" (1 1/8")	**49.478" (4' 1 1/2")**	43.5°
Jack #4	64.0" (5' 4")	67.462" (5' 7 1/2")	1.118" (1 1/8")	**66.344" (5' 6 3/8")**	43.5°
Jack #5	80.0" (6' 8")	84.327" (7' 0 3/8")	1.118" (1 1/8")	**83.209" (6' 11 1/4")**	43.5°

5. Staircase Architecture & IRC Building Code Sentinel

Stair building combines structural geometry with strict safety code compliance and human gait biomechanics. BuildCalc separates legal compliance (IRC dimension limits) from ergonomic comfort (François Blondel stride formula).

5.1 Dual-Mode Solving Algorithm

• **Comfort Mode (Open Space):** Targets the gold-standard ~7.0" riser height: Risers = round(TotalRise / 7.0).
• **Constrained Space Mode (Fixed Stairwell Opening):** Maximizes riser height up to the legal code ceiling (7.75") to minimize horizontal floor footprint: Risers = ceil(TotalRise / 7.75).

5.2 International Residential Code (IRC) Standards

BuildCalc monitors every step configuration against the **Residential Stair Dimension Check — IRC** limits:

• **Maximum Permissible Riser Height: 7 3/4" (7.75 inches / 196 mm)**
• **Minimum Permissible Tread Depth: 10.0 inches (254 mm)** (measured nosing to nosing)
• **Maximum Flight Variation Tolerance: 3/8" (0.375 inches / 9.5 mm)** between greatest and smallest riser or tread

5.3 François Blondel Stride Ratio (1675)

In 1675, French architect François Blondel established the fundamental relationship between human walking stride and vertical stair ascent. Because lifting a foot vertically requires approximately twice the muscular effort of horizontal walking, the sum of two risers and one tread must match a natural walking stride (60 to 64 cm):

```
Blondel Comfort Formula: 2R + T = 24.0" to 25.5" (Ideal: 24.8" / 63 cm)
```

Stair Stringer & Pitch Geometry:

```
Treads = Risers - 1 (Top tread is the finished upper landing)
TotalRun = Treads * TreadDepth
StringerLength = sqrt(TotalRise² + TotalRun²)
InclineAngle = arctan(TotalRise / TotalRun) * (180.0 / π) (Preferred range: 30° to 37°)
```

6. Dimensional Converter & Blueprint Fit Check

Modern construction involves international architectural drawings drafted in metric millimeters alongside off-the-shelf building products manufactured in US Customary nominal inches (e.g., 4x8 drywall sheets, 16" O.C. stud spacing). BuildCalc provides instant bidirectional conversion and a **Cross-Border Fit Check** to identify shimming or trimming variances.

6.1 Blueprint vs. Stock Material Variance Example

Component / Spec	Metric Drawing Spec	Off-the-Shelf Imperial Stock	Dimension Variance	Jobsite Action Required
Standard Sheet Width	1200 mm (3' 11 1/4")	48.0" (1219.2 mm)	`+19.2 mm (+3/4")`	Rip sheet width by 3/4" to fit framing
Ceiling Height / Sheet	2400 mm (7' 10 1/2")	96.0" (2438.4 mm)	`+38.4 mm (+1 1/2")`	Trim sheet ends by 1 1/2" to clear plate
Wall Stud Spacing	600 mm (1' 11 5/8")	24.0" (609.6 mm)	`+9.6 mm (+3/8")`	Layout cumulative drift: 3/8" per bay
Wall Stud Spacing	400 mm (1' 3 3/4")	16.0" (406.4 mm)	`+6.4 mm (+1/4")`	Layout cumulative drift: 1/4" per bay

7. Comprehensive Construction Trade Glossary

Bird's Mouth Notch: A triangular notch cut into the bottom edge of a rafter consisting of a horizontal seat cut resting flat on the top plate and a plumb vertical heel cut resting against the wall frame.

Board Foot (BF): A specialized unit of lumber measurement representing a volume of 144 cubic inches (equivalent to a rough board 1 inch thick, 12 inches wide, and 1 foot long). Calculated using nominal mill dimensions.

Blondel's Ratio: The architectural stride rule 2R + T = 24" - 25.5" formulated by François Blondel in 1675 to ensure human stair climbing cadence is ergonomically natural.

CMU (Concrete Masonry Unit): A standard hollow concrete cinder block. The most common standard 8x8x16 block has manufactured dimensions of 7 5/8" x 7 5/8" x 15 5/8" to fit an exact 8" x 8" x 16" grid when laid with 3/8" mortar.

Common Rafter: A structural roof rafter extending perpendicular from the exterior wall plate to the center ridge beam at the uniform roof pitch angle.

Dry-Stacking: The masonry practice of laying blocks or interlocking retaining stones without mortar joints. Commonly used with rear-lipped landscape retaining walls.

Face-Shell Bedding: The standard trade practice of applying mortar only to the outer horizontal flanges (face shells) and vertical outer edges of hollow CMU blocks, rather than solid full-width bedding.

Hip Rafter: The diagonal structural rafter framing the external junction of two sloping roof planes at an angle (typically 45° in plan view).

Jack Rafter: A shortened rafter that connects an exterior wall plate to a hip rafter (hip jack) or a valley rafter to a ridge beam (valley jack).

On-Center (O.C.) Spacing: The distance measured from the centerline of one structural framing member (stud, joist, or rafter) to the centerline of the adjacent member (standardly 16" or 24" O.C.).

Plumb Cut: Any vertical cut made on a framing timber when installed at its design slope (e.g. at the ridge of a rafter or the tail of a fascia board).

Seat Cut: The horizontal flat cut on a rafter or stair stringer that bears down directly onto a horizontal bearing wall plate or subfloor.

Slump: A standard measure of concrete consistency and fluidity tested with a 12-inch slump cone before pouring. Normal slab slump ranges between 4 and 6 inches.

Stringer: The diagonal structural backbone member supporting the treads and risers of a staircase, cut from 2x12 lumber or fabricated from steel C-channels.

True-Cut Long Point: The physical measuring mark located at the outermost sharp edge of an angled saw bevel cut, ensuring an exact edge-to-plate measurement without centerline conversion math.

8. Verification Matrix & Mathematical Audit Proofs

To guarantee zero regression and verify 100% compliance with mathematical trade standards, the entire application codebase is covered by an automated test suite. All findings from the mathematical audit have been resolved and verified.

Audit Finding	Severity	Root Cause	Engineering Resolution	Test Verification Status
6x6 Board Feet Undercount	**Critical**	Ternary fallback set nominal thickness to 1.0" instead of 6.0" (83% undercount).	Added explicit nominalThicknessInches: 6.0 to NominalLumberSize.	**PASSED: 10 pcs 6x6 @ 12' = 360.0 BF**
2x10 & 2x12 BF Undercount	**Critical**	Ternary fallback defaulted nominal width to 8.0" (20% to 33% undercount).	Added explicit nominalWidthInches (10.0" and 12.0") to enum.	**PASSED: 20 pcs 2x10 @ 16' = 533.33 BF**
CMU Double-Mortar Addition	**Moderate**	Added 0.375" joint to nominal 16.0" creating non-modular 16.375" blocks.	Configured actual manufactured dimensions (15.625" x 7.625") so +0.375" = 16.0" grid.	**PASSED: 30' x 8' wall = 304 blocks w/ 10% waste**
Retaining Wall Inversion & Mortar	**Moderate**	Inverted length/width (8" vs 12") and calculated mortar for dry-stacked stones.	Set length=12", height=4", depth=8" and added isDryStacked: true (0 mortar).	**PASSED: 10' x 3' wall = 99 blocks, 0.0 mortar bags**
Ready-Mix Truck Rounding	**Norm**	Rounded orders to 0.1 yd^3; batch plants standardly batch quarter-yards.	Updated recommendedOrderYards to ceil(gross * 4.0) / 4.0.	**PASSED: 3.10 yd^3 gross orders 3.25 yd^3**
Converter Decimal Feet Truncation	**Edge-Case**	val.toInt() stripped decimal portion when typing 12.5 ft.	Updated conversion formula to (val * 12.0) + fraction.	**PASSED: 12.5 ft preserves exact 150.0" (12' 6")**

8.2 Automated Test Execution Summary

Test Suite Name	Test File Path	Tests Executed	Pass Rate
Fraction Engine Tests	`test/core/math/fraction_test.dart`	4 tests	100% (4 / 4)
Stair, Concrete, Framing & Masonry	`test/core/math/stair_and_concrete_test.dart`	15 tests	100% (15 / 15)
Roof Diagram & Common Rafters	`test/features/roofing/roof_diagram_test.dart`	2 tests	100% (2 / 2)
Progressive Disclosure & Guides	`test/progressive_disclosure_test.dart`	1 test (multistep)	100% (1 / 1)
App Launch & Shell Navigation	`test/widget_test.dart`	2 tests	100% (2 / 2)
Multi-Viewport Responsive QA (320-430px)	`test/responsive_viewport_test.dart`	27 tests	100% (27 / 27)
Total Test Suite Coverage	`All core modules verified`	**51 tests total**	**100% PASS (51 / 51)**

TECHNICAL CERTIFICATION & PRODUCTION SIGN-OFF

This technical specification confirms that BuildCalc™ v1.0 meets all structural mathematics, building code thresholds, and fieldwork trade norms documented herein. All mathematical algorithms, dimensional formulas, unit conversions, and responsive user interface components have been verified via automated regression suites with zero errors and zero layout overflows.

Status: PRODUCTION READY & CODE COMPLIANT